Training
Report
on
"MOVIE RECOMMENDATION SYSTEM"

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

Under the

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CANDIDATE'S DECLARATION

I hereby declare that the work presented in this training report entitled "**Predictive Analysis of Student Academic Performance Using Linear Regression**", submitted in partial fulfilments of the requirement for the award of the degree of **Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence and Machine Learning)**, submitted to **Panipat Institute of Engineering & Technology, Kurukshetra University, Kurukshetra, India** is an authentic record of my own work carried out during the period from 21/07/2024 to 25/08/2024 under the guidance of Mr. Deepak Singla.

The work reported in this project report has not been submitted by me for the award of any other degree or diploma.

Date: 13-09-2025 **Submitted By :-**

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CHAPTER 1
INTERODUCTION

The rapid growth of digital entertainment platforms has significantly transformed the way people discover and consume movies. With thousands of films available online, selecting a movie that aligns with a viewer's personal taste has become increasingly challenging. To address this problem, recommendation systems play a vital role by analyzing user preferences and suggesting relevant movies.

This project, "Movie Recommendation System Using Python," focuses on providing movie suggestions based on the user's input. By utilizing Python libraries such as Pandas, NumPy, and Matplotlib, as well as a comprehensive dataset obtained from Kaggle, this system analyzes patterns within the movie dataset to identify and recommend similar movies.

The project involves essential steps such as data collection, cleaning, preprocessing, feature extraction, and the implementation of a similarity-based recommendation approach. Pandas is used for effective data handling and analysis, NumPy supports numerical operations, and Matplotlib helps in visualizing data trends such as genre distribution and movie popularity.

This introductory chapter outlines the motivation behind the system, the need for recommendations in today's digital era, and the importance of data-driven decision-making. It further highlights the project's objective—to develop a simple yet efficient movie suggestion system that enhances user experience and demonstrates fundamental concepts of data analysis and Python programming.

Overall, this project not only provides practical exposure to handling real-world datasets but also strengthens the understanding of Python libraries, data processing, and visualization techniques essential for building intelligent applications.

COMPANY PROFILE

BrainHub is a technology-driven organization focused on delivering innovative digital solutions, advanced software development services, and AI-powered tools to meet modern industry demands. The company is known for its strong emphasis on research, creativity, and problem-solving, enabling businesses and learners to explore the true potential of emerging technologies.

BrainHub provides a collaborative learning environment and hands-on project exposure, helping students, interns, and professionals enhance their skills in fields like **Artificial Intelligence, Machine Learning, Data Science, Full Stack Development, and Cloud Technologies**.

INTERNSHIP OBJECTIVES

Purpose

The objective of this internship is to gain practical, hands-on experience in applying theoretical knowledge to real-world projects. It serves as a bridge between academic learning and professional practice.

Key Goals

- **Skill Development:** Enhance technical skills in programming, data analysis, and system optimization.
- **Industry Exposure:** Understand workflows, tools, and methodologies used in professional environments.
- **Problem-Solving:** Apply critical thinking and modular approaches to tackle complex challenges.
- **Collaboration:** Work effectively in teams, contributing to shared goals and learning from peers and mentors.
- **Communication:** Improve ability to document, present, and explain technical concepts clearly.
-

Expected Outcomes

- Build confidence in applying AI/ML and software engineering concepts to practical problems.
- Develop a portfolio of projects that demonstrate technical and strategic capabilities.
- Gain insights into career pathways and professional expectations.
- Strengthen adaptability and resilience in high-pressure, competitive environments.

WORK DONE

Modern OTT platforms such as Netflix and Amazon Prime use recommendation systems to help users easily discover movies they would like. The volume of films available online is huge, making it difficult for users to manually find movies based on their tastes. To solve this issue, we created a Python-based Movie Suggestion System.
Our system uses a dataset downloaded from Kaggle, analyzes it using Pandas, performs numerical operations using NumPy, and visualizes data trends using Matplotlib.

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Long-Term Vision

This internship is a stepping stone toward becoming a product-focused technologist who combines technical precision with strategic communication, aiming to deliver impactful solutions that empower others.

Overview

During the internship period, I have engaged in multiple projects and tasks that have enhanced my technical and professional skills. The work done spans across software development, data analysis, and collaborative team efforts.

Projects and Contributions

1. AI-Based Movie Recommender System

- Developed a Python-based movie recommendation engine using collaborative filtering techniques.
- Implemented data preprocessing, similarity calculations, and user interface integration.
- Optimized the system for performance and accuracy, achieving a significant improvement in recommendation relevance.

2. Digital System Design Modules

- Designed and simulated various digital circuits including flip-flops, counters, and multiplexers.
- Created modular, reusable components to facilitate rapid prototyping and testing.
- Documented design processes and results with clear diagrams and explanations.

3. Data Structures and Algorithms

- Implemented core data structures such as linked lists, stacks, queues, and trees in C++.
- Developed algorithms for sorting, searching, and graph

traversal.
- Analyzed time and space complexity to optimize code efficiency.

4. Collaborative Team Projects

- Participated in team-based hackathons focusing on AR applications for tourism.
- Contributed to strategic planning, coding, and presentation preparation.
- Enhanced communication and teamwork skills through regular meetings and feedback sessions.

Skills Developed

- Proficient in Python, C++, and C programming languages.
- Strong understanding of AI/ML concepts and their practical applications.
- Ability to modularize complex workflows for rapid development.
- Effective collaboration and communication within diverse teams.
- Problem-solving under tight deadlines and competitive environments.

Reflections and Learnings

This internship has been instrumental in bridging the gap between academic theory and real-world application. The hands-on experience has solidified my understanding of core concepts and boosted my confidence in tackling complex technical challenges. The collaborative environment fostered growth in interpersonal skills and adaptability, preparing me for future professional endeavors.

Future Directions

Building on this foundation, I aim to further explore product development with a focus on user-centric design and scalable solutions. Continuing to refine my technical expertise and strategic communication will be key to achieving long-term career goals in technology and innovation.# Internship Objective

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CODING EXPLANATION

1. Importing Libraries

```
import pandas as pd
import ast
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics.pairwise import cosine_similarity
```

- **pandas**: For handling datasets.
- **ast**: To safely evaluate strings containing Python objects (like lists/dictionaries).
- **CountVectorizer**: Converts text data into numerical vectors.
- **cosine_similarity**: Measures similarity between vectors.

---

2. Loading Datasets

```
movies = pd.read_csv("tmdb_5000_movies.csv")
credits = pd.read_csv("tmdb_5000_credits.csv")
```

- Loads movie and credits datasets from CSV files.

---

3. Merging and Selecting Columns

```
movies = movies.merge(credits, on='title')
movies = movies[['movie_id', 'title', 'overview', 'genres', 'keywords', 'cast',
'crew']]
```

- Merges datasets on **title**.
- Keeps only relevant columns.

---

4. Helper Functions

```
def extract_names(obj):
    return [i['name'] for i in ast.literal_eval(obj)] if pd.notnull(obj) else []

def extract_cast(obj):
    return [i['name'] for i in ast.literal_eval(obj)[:3]] if pd.notnull(obj) else []

def extract_director(obj):
    return [i['name'] for i in ast.literal_eval(obj) if i['job']=='Director'] if
pd.notnull(obj) else []
```

- **extract_names**: Extracts names from JSON-like strings.

- **extract_cast**: Extracts top 3 cast members.
- **extract_director**: Extracts director name.

---

5. Applying Transformations

```
movies['genres']   = movies['genres'].apply(extract_names)
movies['keywords'] = movies['keywords'].apply(extract_names)
movies['cast']     = movies['cast'].apply(extract_cast)
movies['crew']     = movies['crew'].apply(extract_director)
movies['overview'] = movies['overview'].apply(lambda x: x.split() if
isinstance(x,str) else [])
```

- Converts text into lists of words.
- Extracts genres, keywords, cast, and crew.

---

6. Creating Tags

```
movies['tags'] = movies['overview'] + movies['genres'] + movies['keywords'] +
movies['cast'] + movies['crew']
new_df = movies[['movie_id','title','tags']]
new_df['tags'] = new_df['tags'].apply(lambda x: " ".join(x).lower())
```

- Combines all features into a single **tags** column.
- Converts tags into lowercase strings.

---

7. Vectorization and Similarity

```
cv = CountVectorizer(max_features=5000, stop_words='english')
vectors = cv.fit_transform(new_df['tags']).toarray()
similarity = cosine_similarity(vectors)
```

- **CountVectorizer**: Converts tags into numerical vectors.
- **cosine_similarity**: Creates similarity matrix between movies.

---

8. Recommendation Function

```
def recommend(movie):
    if movie not in new_df['title'].values:
        print(f"Movie '{movie}' not found.")
        return
    idx = new_df[new_df['title']==movie].index[0]
    distances = sorted(list(enumerate(similarity[idx])), key=lambda x:x[1],
reverse=True)
    print(f"\nMovies similar to '{movie}':")
```

```
for i in distances[1:6]:
    print(new_df.iloc[i[0]].title)
```

- Finds the index of the given movie.
- Sorts movies by similarity score.
- Prints top 5 similar movies.

---

9. Main Program

```
movie_name = input("Enter a movie name: ")
recommend(movie_name.strip())
```

- Takes user input.
- Calls the recommendation function.

Summary

This code builds a **content-based movie recommendation system** using Python. It:

- Loads and preprocesses data.
- Extracts features like genres, cast, and keywords.
- Creates tags and converts them int

CONCLUSION

This project demonstrates how a recommendation system can be built using only Python and its basic data-handling libraries — Pandas, NumPy, and Matplotlib. Even without advanced machine learning libraries, the system successfully analyzes user input and provides relevant movie suggestions. The project strengthens understanding of data preprocessing, similarity calculations, and visualization.

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THANKS GIVING

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This project has been a valuable learning experience, and we sincerely thank everyone who contributed directly or indirectly to its successful completion.

ANNEXURES

Annexure I: Code Overview

Python-based movie recommendation system using cosine similarity.

Utilizes TMDB 5000 Movies and Credits datasets.

Implements manual Bag-of-Words vectorization.

Annexure II: Data Processing Steps

Load datasets (movies and credits).

Merge datasets on movie title.

Extract required fields (genres, keywords, cast, crew).

Convert nested JSON-like strings using ast.literal_eval.

Generate tags combining overview + metadata.

Create feature vectors.

Annexure III: Recommendation Logic

Cosine similarity between vectors.

Returns top 5 similar movies.

Annexure IV: Dependencies

pandas

numpy

ast

matplotlib